CUSTOMER CHURN ANALYSIS AND RETENTION STRATEGY

A Data-Driven Business Intelligence Project

Technical Report

Project Domain: Data Analytics and Business Intelligence

Tools and Technologies:

- Python
- Pandas and NumPy
- Matplotlib and Seaborn
- SciPy
- Jupyter Notebook

Dataset: Customer Churn Dataset (500 customer records)

Abstract

Customer churn is an important business challenge because losing existing customers can affect revenue, customer relationships, and long-term business growth.

This project focuses on analysing customer churn using Python and statistical techniques. The dataset contains 500 customer records with information about tenure, monthly charges, total charges, contract type, payment method, paperless billing, and churn status.

The analysis includes data cleaning, exploratory data analysis (EDA), customer segmentation, and hypothesis testing. The dataset shows an overall churn rate of 10.6% and a customer retention rate of 89.4%.

Statistical testing identified significant differences in average tenure and monthly charges between retained and churned customers. It also identified a significant association between contract type and customer churn.

Based on these findings, the project recommends improving customer onboarding, reviewing pricing strategies, and developing targeted retention programmes.

Keywords: Customer Churn, Data Analytics, Python, Exploratory Data Analysis, Hypothesis Testing, Customer Retention, Business Intelligence.

1. INTRODUCTION

1.1 Background

Customer churn refers to the situation in which customers stop using a company's products or services. It is an important business problem because retaining existing customers is essential for maintaining stable revenue and building long-term customer relationships.

Understanding customer behaviour helps businesses identify customers who are more likely to leave. Data analytics provides methods to examine customer information, identify patterns, and develop effective retention strategies.

This project uses Python-based data analytics techniques to study customer churn and identify important factors associated with customer decisions to leave a service.

1.2 Problem Statement

Businesses may experience customer loss without having a clear understanding of the factors associated with churn. Without analysing customer data, it becomes difficult to identify high-risk customer groups and develop suitable retention strategies.

Therefore, this project aims to analyse customer data, identify patterns related to churn, and provide data-driven recommendations to improve customer retention.

1.3 Project Objectives

The main objectives of this project are:

1. To examine and understand the customer churn dataset.
2. To perform data cleaning and validation.
3. To calculate important business KPIs, including churn rate and retention rate.
4. To perform exploratory data analysis using Python.
5. To analyse the relationship between customer attributes and churn.
6. To conduct statistical hypothesis testing.
7. To identify customer groups with higher observed churn rates.
8. To recommend strategies for improving customer retention.

1.4 Scope of the Project

This project focuses on analysing a dataset containing 500 customer records and 9 attributes.

The analysis covers:

- Customer tenure and monthly charges.
- Total customer charges.
- Contract types.
- Payment methods.
- Paperless billing.
- Customer churn status.

The project uses descriptive statistics, data visualization, and statistical hypothesis testing to identify patterns and associations.

The recommendations are based on the available dataset. The project does not establish causal relationships or predict future churn using a machine learning model.

1.5 Significance of the Project

The project demonstrates how data analytics can support business decision-making. By identifying customer groups with higher churn rates, businesses can develop targeted retention strategies, improve customer experiences, and make more informed decisions.

2. DATASET DESCRIPTION AND DATA CLEANING

2.1 Dataset Overview

The project uses a customer churn dataset containing 500 customer records and 9 columns. Each record represents an individual customer.

The dataset includes customer information such as tenure, monthly charges, total charges, contract type, payment method, paperless billing, and churn status.

Dataset Attributes

Attribute	Description	Data Type
CustomerID	Unique customer identifier	String
Tenure	Number of months the customer has stayed	Integer
MonthlyCharges	Monthly amount charged to the customer	Integer
TotalCharges	Total amount charged to the customer	Integer
Contract	Customer's contract category	Categorical
PaymentMethod	Customer's payment method	Categorical
PaperlessBilling	Indicates paperless billing status	Categorical
SeniorCitizen	Senior citizen indicator	Integer
Churn	Customer churn status (0 or 1)	Integer

2.2 Tools and Technologies Used

The following tools and libraries were used:

- **Python:** Main programming language for data analysis.
- **Pandas:** Data loading, manipulation, grouping, and analysis.
- **NumPy:** Numerical operations.

- **Matplotlib:** Data visualization.
- **Seaborn:** Statistical charts and visualizations.
- **SciPy:** Statistical hypothesis testing.
- **Jupyter Notebook:** Interactive development and documentation environment.

2.3 Data Cleaning Methodology

Data cleaning was performed using Pandas to check the quality and consistency of the dataset.

The following checks were conducted:

1. **Dataset inspection:** Checked the number of rows, columns, and data types.
2. **Missing value analysis:** Checked every column for missing values.
3. **Duplicate record detection:** Checked for duplicate rows.
4. **Customer ID validation:** Verified that customer identifiers were unique.
5. **Categorical value inspection:** Examined categories for inconsistent labels.
6. **Numerical range inspection:** Reviewed the minimum and maximum values of numerical attributes.

2.4 Data Cleaning Results

The following results were obtained:

Data Quality Check	Result
Total records	500
Total columns	9
Missing values	0
Duplicate rows	0
Duplicate Customer IDs	0
Unique Customer IDs	500

No missing values, duplicate records, or inconsistent categorical labels were identified during the initial validation.

Therefore, no major data corrections were required. A validated copy of the original dataset was exported as cleaned_data.csv.

2.5 Conclusion

The dataset passed the initial data quality and validation checks. The cleaned dataset was used for further exploratory data analysis and statistical investigation.

3. EXPLORATORY DATA ANALYSIS (EDA)

3.1 Introduction

Exploratory Data Analysis (EDA) was performed to understand customer behaviour, identify patterns, and examine the factors associated with customer churn.

Pandas, Matplotlib, and Seaborn were used to calculate business KPIs and create visualizations.

3.2 Key Performance Indicators (KPIs)

The following KPIs were calculated:

KPI	Value
Total Customers	500
Retained Customers	447
Churned Customers	53
Retention Rate	89.4%
Churn Rate	10.6%
Average Tenure	36.53 months

The results indicate that most customers remained with the company, while 10.6% discontinued the service.

3.3 Customer Churn Distribution

The dataset contains 447 retained customers and 53 churned customers.

- Retained customers: 89.4%
- Churned customers: 10.6%

This indicates that the majority of customers continued using the service.

Business Insight: Although the overall churn rate is relatively low, understanding the characteristics of the customers who leave can help improve retention strategies.

3.4 Customer Tenure Analysis

Customer tenure represents the number of months a customer has stayed with the company.

The average customer tenure was 36.53 months, and the median tenure was 37 months.

The tenure distribution showed the following:

Tenure Range (Months)	Number of Customers
0–15	110
16–30	94
31–45	99
46–60	118
61–100	79

The largest group of customers had a tenure between 46 and 60 months.

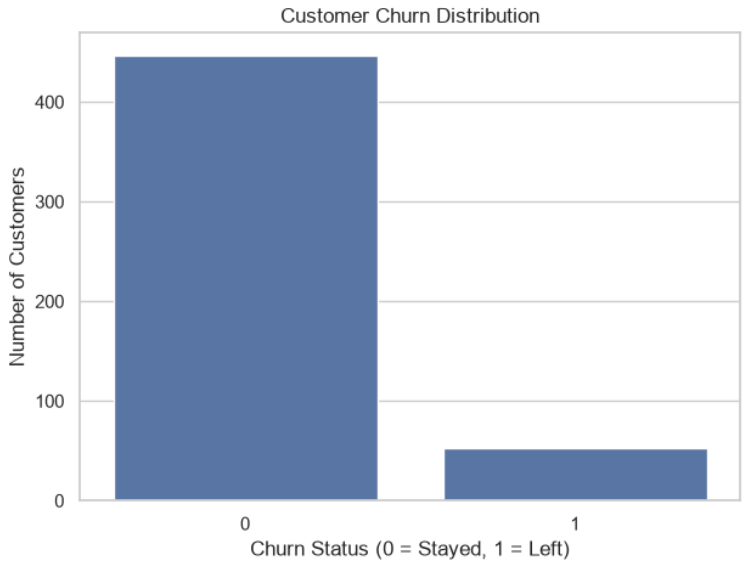


Fig.1 Customer Churn Distribution

Business Insight: Customer tenure provides useful information about customer loyalty and retention patterns.

3.5 Contract Type Analysis

The dataset contains three contract categories:

Contract Type	Customers
One Year	186
Month-to-Month	170
Two Year	144

One-year contracts were the most common contract type.

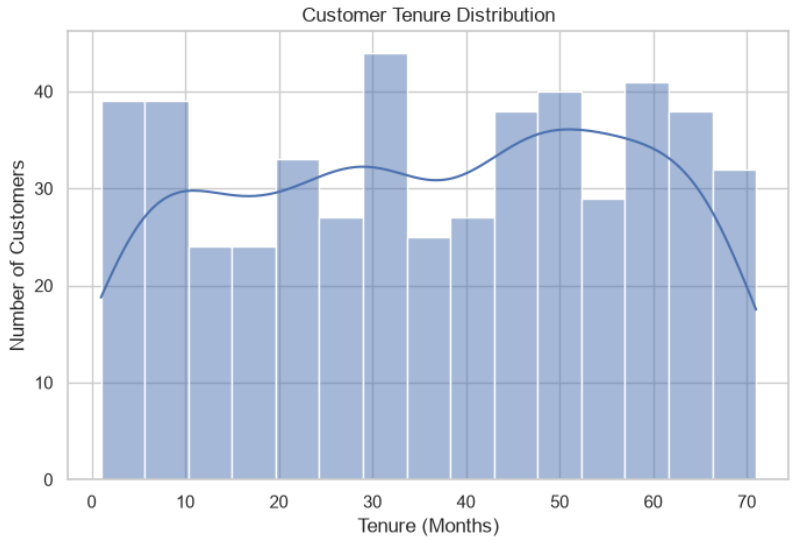


Fig 2. Customer Tenure Distribution

The observed churn rates by contract type were:

Contract Type	Churn Rate
Month-to-Month	20.59%
One Year	4.30%
Two Year	6.94%

Month-to-month customers had the highest observed churn rate.

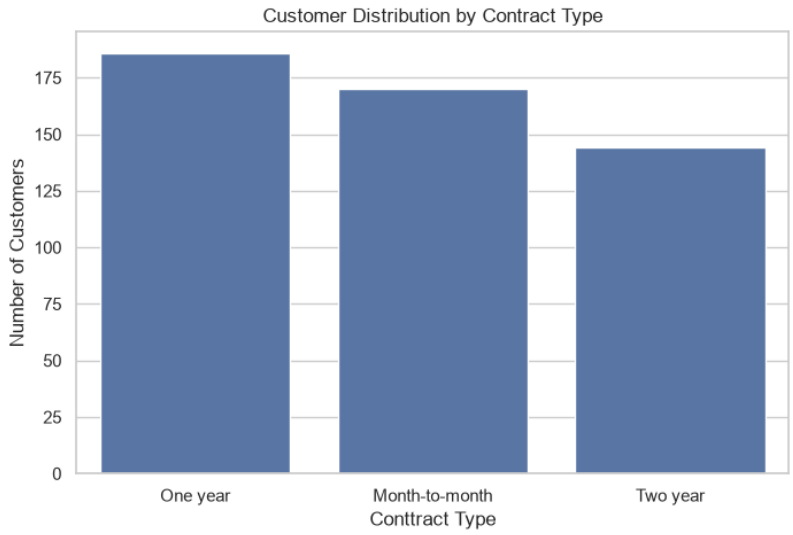


Fig 3. Customer Distribution by Contract Type

Business Insight: Businesses could consider suitable loyalty benefits and incentives to encourage customers to explore longer-term plans.

3.6 Payment Method Analysis

The churn rates across payment methods were:

Payment Method	Churn Rate
Bank Transfer	6.92%
Credit Card	13.48%
Electronic Check	11.04%

Customers using credit cards had the highest observed churn rate among the three payment methods.

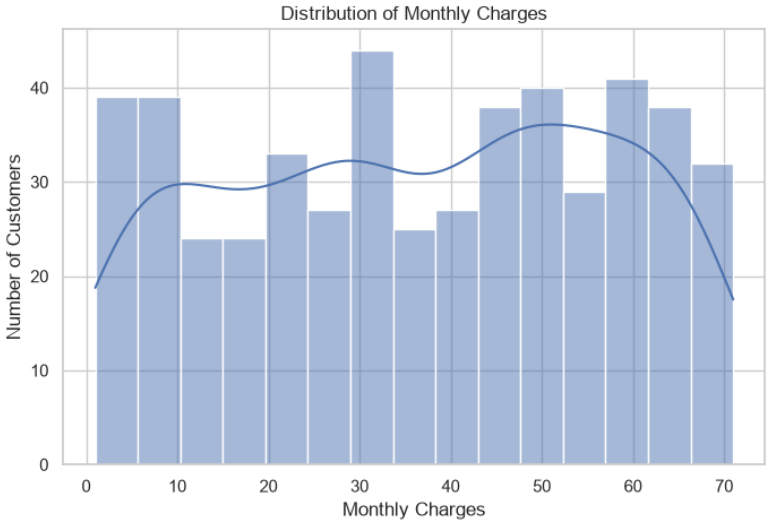


Fig 4. Distribution of Monthly Charges

Business Insight: The company could investigate customer experiences across payment methods to understand whether any payment-related issues contribute to dissatisfaction.

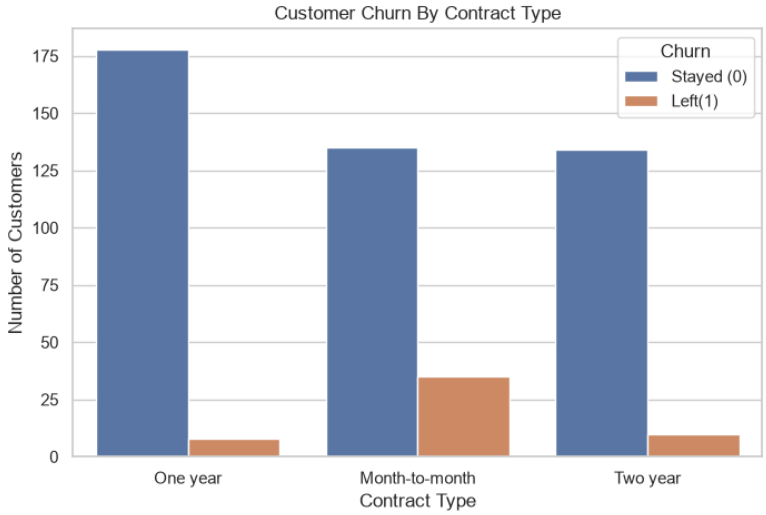


Fig 5. Customer Churn By Contract Type

3.7 Paperless Billing Analysis

The churn rates based on paperless billing status were:

Paperless Billing	Churn Rate
No	10.12%
Yes	11.11%

The difference between the two groups was small.

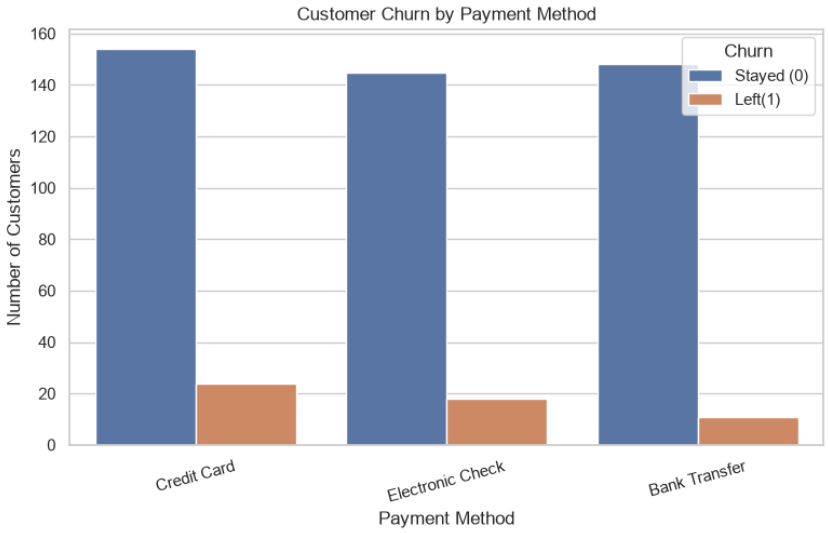


Fig 6. Customer Churn by Payment Method

Business Insight: Paperless billing showed only a marginal difference in observed churn rates and should not be treated as a major churn indicator based on this analysis alone.

3.8 Monthly Charges Analysis

The average monthly charges were compared between retained and churned customers.

Customer Status	**Average Monthly Charges**
Retained	111.72
Churned	129.77

Churned customers had approximately 18.05 higher average monthly charges than retained customers.

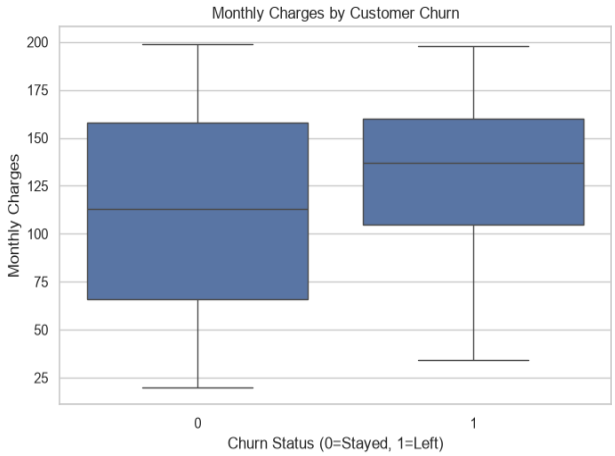


Fig 7. Monthly Charges by Customer Churn

Business Insight: The company could review pricing plans and investigate whether higher charges are associated with customer dissatisfaction.

3.9 Correlation Analysis

Correlation analysis was performed to examine the relationships between numerical attributes and customer churn.

The correlation between tenure and churn was approximately -0.51, indicating a moderate negative relationship.

The correlation between monthly charges and churn was approximately 0.11, indicating a very weak positive relationship. Total charges had a correlation of approximately 0.00 with churn.

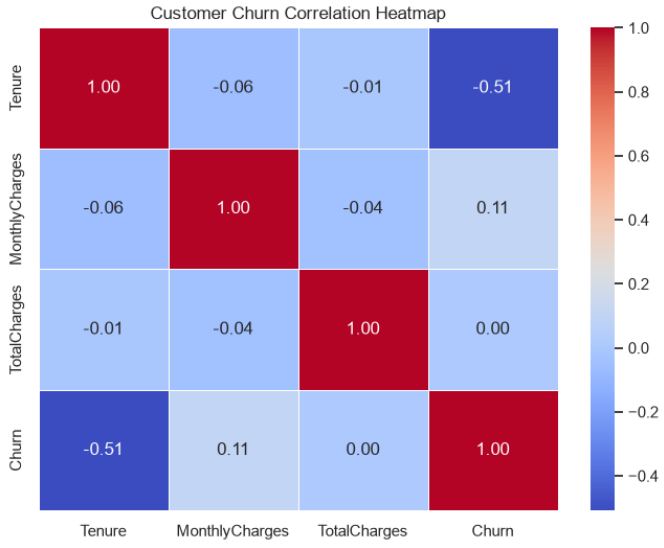


Fig 8. Customer Churn Correlation Heatmap

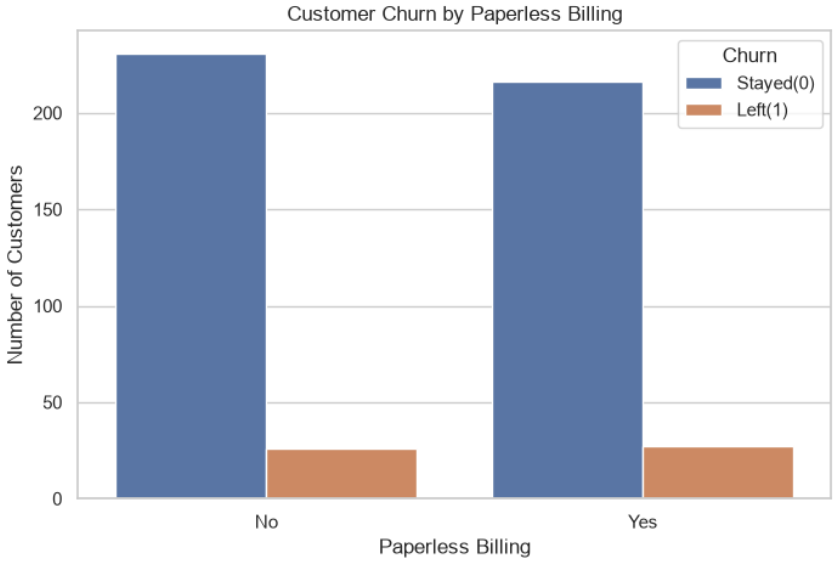


Fig 9. Customer Churn by Paperless Billing

Business Insight: Tenure showed the strongest numerical correlation with churn among the variables examined. However, correlation does not establish causation.

3.10 EDA Conclusion

The exploratory analysis identified several important patterns:

1. The overall customer churn rate was 10.6%.
2. Churned customers had substantially lower average tenure.
3. Month-to-month customers had the highest observed churn rate.
4. Churned customers had higher average monthly charges.
5. Tenure showed the strongest numerical correlation with churn.

These findings provide a foundation for the statistical analysis and business recommendations presented in the following sections.

4. STATISTICAL ANALYSIS

4.1 Introduction

Statistical analysis was performed to determine whether selected customer attributes have statistically significant differences or associations with customer churn.

Three hypothesis tests were conducted using the SciPy library.

The significance level was set to 0.05.

4.2 Hypothesis Testing Methodology

Hypothesis testing was used to evaluate the relationships between customer attributes and churn.

The following decision rules were applied:

- If the p-value is less than 0.05, reject the null hypothesis (H0).
- If the p-value is greater than or equal to 0.05, fail to reject the null hypothesis (H0).

The tests selected were:

1. Welch's Independent Samples t-test.
2. Chi-square Test of Independence.

4.3 Test 1: Tenure vs Churn

Objective: To determine whether average customer tenure differs significantly between retained and churned customers.

Null Hypothesis (H0): There is no significant difference in average tenure between retained and churned customers.

Alternative Hypothesis (H1): There is a significant difference in average tenure between retained and churned customers.

Test Used: Welch's Independent Samples t-test.

Results:

- Average tenure of retained customers: 40.15 months.

- Average tenure of churned customers: 6 months.
- P-value: 8.982326e-130.
- Decision: Reject H0.

Interpretation:

The p-value is below 0.05, indicating a statistically significant difference in average tenure between the two groups.

Churned customers had substantially lower average tenure than retained customers.

Business Implication: The company should focus on improving onboarding, customer support, and engagement during the early stages of the customer relationship.

4.4 Test 2: Monthly Charges vs Churn

Objective: To determine whether average monthly charges differ significantly between retained and churned customers.

Null Hypothesis (H0): There is no significant difference in average monthly charges between retained and churned customers.

Alternative Hypothesis (H1): There is a significant difference in average monthly charges between retained and churned customers.

Test Used: Welch's Independent Samples t-test.

Results:

- Average monthly charges of retained customers: 111.72.
- Average monthly charges of churned customers: 129.77.
- P-value: 0.0100792.
- Decision: Reject H0.

Interpretation:

The p-value is below 0.05, indicating a statistically significant difference in average monthly charges between retained and churned customers.

Churned customers had higher average monthly charges.

Business Implication: The company should review its pricing structure and consider suitable pricing options and targeted retention offers.

4.5 Test 3: Contract Type vs Churn

Objective: To determine whether contract type is statistically associated with customer churn.

Null Hypothesis (H0): Contract type and customer churn are independent.

Alternative Hypothesis (H1): Contract type and customer churn are associated.

Test Used: Chi-square Test of Independence.

Results:

- Chi-square statistic: 27.7153.
- Degrees of freedom: 2.
- P-value: 9.587354e-07.
- Decision: Reject H0.

Interpretation:

The p-value is below 0.05, indicating a statistically significant association between contract type and customer churn.

Month-to-month customers had the highest observed churn rate of 20.59%.

Business Implication: The company could introduce suitable loyalty programmes and incentives to encourage customers to consider longer-term contracts.

4.6 Summary of Hypothesis Testing

Analysis	Statistical Test	P-value	Decision
Tenure vs Churn	Welch's t-test	8.982326e-130	Reject H0
Monthly Charges vs Churn	Welch's t-test	0.0100792	Reject H0
Contract Type vs Churn	Chi-square test	9.587354e-07	Reject H0

All three tests produced p-values below the significance level of 0.05.

Therefore, the analysis found statistically significant differences in tenure and monthly charges, and a statistically significant association between contract type and churn.

4.7 Statistical Analysis Conclusion

The hypothesis testing results provide statistical evidence that customer tenure and monthly charges differ between retained and churned customers. They also show that contract type is associated with customer churn.

These findings support the development of targeted customer retention strategies.

However, statistical significance does not establish causation. Further investigation would be required to understand the underlying reasons for customer churn.

5. BUSINESS RECOMMENDATIONS

5.1 Introduction

Based on the exploratory data analysis and statistical hypothesis testing, several customer retention opportunities were identified.

The following recommendations are designed to help businesses understand customer behaviour, reduce churn risk, and improve customer retention.

5.2 Improve Customer Onboarding

Finding: Churned customers had an average tenure of only 6 months, compared with 40.15 months for retained customers.

Recommendation:

- Introduce a structured onboarding programme.
- Provide guidance during the first few months.
- Offer responsive customer support.
- Collect early feedback to identify customer concerns.

Expected Business Benefit: Improved early customer engagement and reduced early-stage churn risk.

5.3 Review Pricing Strategies

Finding: Churned customers had higher average monthly charges (129.77) compared with retained customers (111.72).

Recommendation:

- Review existing pricing plans.
- Introduce flexible subscription options.
- Consider targeted discounts for eligible customers.
- Communicate the value of services clearly.

Expected Business Benefit: Improved perceived value and potentially better customer retention.

5.4 Encourage Long-Term Contracts

Finding: Month-to-month customers had the highest observed churn rate of 20.59%.

Recommendation:

- Introduce loyalty rewards.
- Offer suitable benefits for longer-term contracts.
- Provide incentives for eligible customers to renew.
- Explain the benefits of different contract options.

Expected Business Benefit: Encourage customer loyalty and reduce churn among high-risk customer groups.

5.5 Investigate Payment-Related Customer Experience

Finding: Customers using credit cards had the highest observed churn rate among the payment methods analysed (13.48%).

Recommendation:

- Investigate customer feedback related to payment experiences.
- Check whether billing issues or payment difficulties are present.
- Make payment processes convenient and transparent.

Expected Business Benefit: Identify and address possible payment-related customer concerns.

5.6 Develop Customer Retention Monitoring

Recommendation:

- Monitor churn rate regularly.
- Track customer tenure and contract distribution.
- Compare retention across payment methods.
- Review customer feedback.

- Evaluate the performance of retention initiatives.

Expected Business Benefit: Enable continuous monitoring and timely identification of customer retention issues.

5.7 Summary of Recommendations

Identified Area	**Recommended Action**
Low tenure among churned customers	Improve onboarding and early support
Higher average monthly charges	Review pricing and flexible plans
High churn among month-to-month customers	Introduce loyalty programmes
Higher observed churn among credit card users	Investigate payment experiences
Need for ongoing monitoring	Track customer retention KPIs

5.8 Recommendation Limitations

These recommendations are based on observed patterns and statistical associations in the available dataset.

The analysis does not establish that any individual attribute causes customer churn. The company should validate these recommendations through customer feedback, further investigation, and controlled retention experiments.

6. CONCLUSION, LIMITATIONS AND FUTURE SCOPE

6.1 Conclusion

The Customer Churn Analysis and Retention Strategy project successfully demonstrated how Python and statistical techniques can be used to analyse customer behaviour and identify patterns associated with churn.

The project analysed 500 customer records containing 9 attributes. Data cleaning and validation confirmed that the dataset had no missing values or duplicate records.

Exploratory Data Analysis identified an overall churn rate of 10.6% and a customer retention rate of 89.4%.

The analysis revealed several important findings:

1. Churned customers had a significantly lower average tenure of 6 months compared with 40.15 months for retained customers.
2. Churned customers had higher average monthly charges of 129.77 compared with 111.72 for retained customers.
3. Month-to-month customers had the highest observed churn rate of 20.59%.
4. Tenure showed the strongest numerical correlation with churn among the numerical variables examined.
5. Statistical testing identified significant differences in tenure and monthly charges, along with a significant association between contract type and churn.

Based on these findings, the project recommended improving customer onboarding, reviewing pricing strategies, introducing suitable loyalty programmes, and monitoring customer retention KPIs.

Overall, the project demonstrates the importance of data-driven decision-making in understanding customer churn and developing customer retention strategies.

6.2 Project Limitations

The project has the following limitations:

1. **Dataset Size:** The analysis was performed on 500 customer records, which may not represent the entire customer population.
2. **Limited Attributes:** The dataset does not include detailed customer feedback, reasons for cancellation, service complaints, or customer satisfaction scores.
3. **No Causal Analysis:** Statistical associations do not prove that the identified attributes cause customer churn.
4. **No Predictive Model:** The project focuses on descriptive and statistical analysis rather than predicting future customer churn.
5. **Limited Time Information:** The dataset does not provide detailed historical customer behaviour over time.

6.3 Future Scope

The project can be extended in several ways:

1. **Machine Learning:** Develop classification models such as Logistic Regression, Decision Trees, and Random Forest to predict customer churn.
2. **Customer Segmentation:** Group customers based on tenure, spending patterns, and contract types.
3. **Interactive Dashboard:** Develop a dashboard using Power BI, Tableau, or Streamlit to monitor customer churn KPIs.
4. **Advanced Analytics:** Explore additional statistical methods to understand customer behaviour.
5. **Retention Campaign Evaluation:** Measure the effectiveness of customer retention programmes using controlled experiments.
6. **Automated Monitoring:** Build a system to regularly monitor customer churn indicators and generate business insights.

6.4 Final Remarks

This project provided practical experience in data cleaning, exploratory data analysis, statistical hypothesis testing, data visualization, and business interpretation using Python.

It also demonstrated how analytical findings can be converted into practical recommendations to support business decisions and improve customer retention.